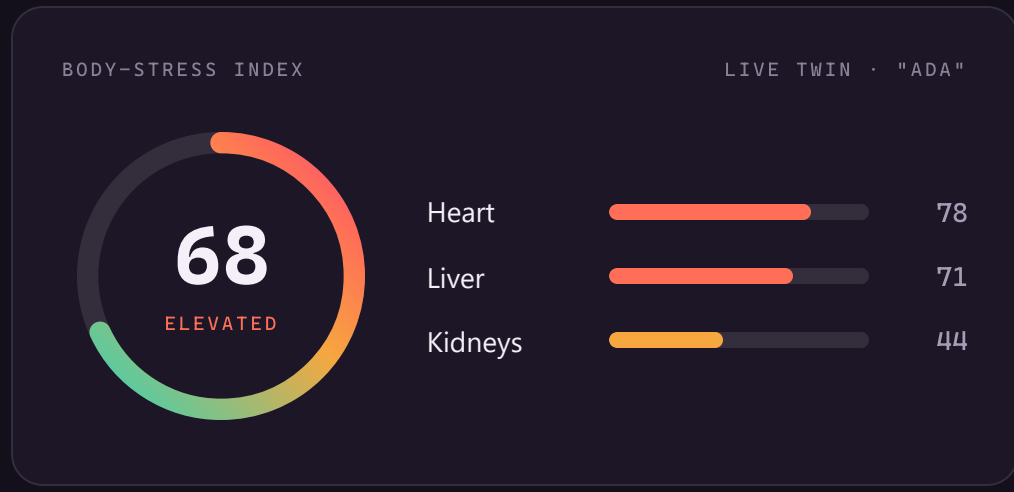
ONTOMORPH HACKATHON · OAU · 2026

Twinstate

A living health twin for students. Log what you're taking — it builds a 3D body that **visibly reacts**, screens the combination against real clinical knowledge, and explains the risk in plain language, with a clear path to care.

- Powered by HOLON
- 3D anatomy
- AI reasoning



EXAM-SEASON POLYPHARMACY

Nobody checks the combination.

Each thing feels harmless alone. Stacked together through exam season, they quietly strain the heart, liver, kidneys and gut — and no one is looking at the total load.

- energy drinks
- coffee
- paracetamol
- ibuprofen
- "study aids"

~400 mg/day

Caffeine ceiling — a heavy exam day sails past it without a thought.

~4 g/day

Paracetamol ceiling — easy to repeat-dose over when you feel rough.

Twinstate is the missing safety check — made **visual** and **personal**, for the student first and the campus health centre that sees the fallout second.

FROM A SCATTERED STACK TO A BODY YOU CAN READ

See what the stack is doing to you.

01 Log it — under two minutes

Rough doses are fine. Anything you type resolves to a real clinical concept, even off-catalogue names like "Augmentin".

02 The twin reacts

A live 3D body recolours organ by organ — a beating heart, a scanning beam — driven directly by your risk data.

03 HOLON screens the whole list

The full combination is checked for interactions: severity, mechanism, clinical effect, management and evidence grade.

04 Plain language + a path to care

Graded calm → watch → caution → urgent, in a friend's voice — never "stop your meds", always "here's your next step".

One living body, end to end — five pieces of the platform: Digital-twin model · 3D anatomy · HOLON knowledge · AI reasoning · Simulation

NOT A CHECKER — A LIVING TWIN

Being told a risk vs. watching it.

- **A living 3D body** that reacts to your own inputs — the data model you can see, not a text box and a list.
- **Two projected futures** side by side — "keep this pattern" vs "rest + hydrate" — so the trade-off is something you watch.
- **AI computes your numbers** from your data; interactions come from HOLON, never invented by the model.
- **Grounded health chat** with a hard safety rule — anything urgent routes straight to real care.

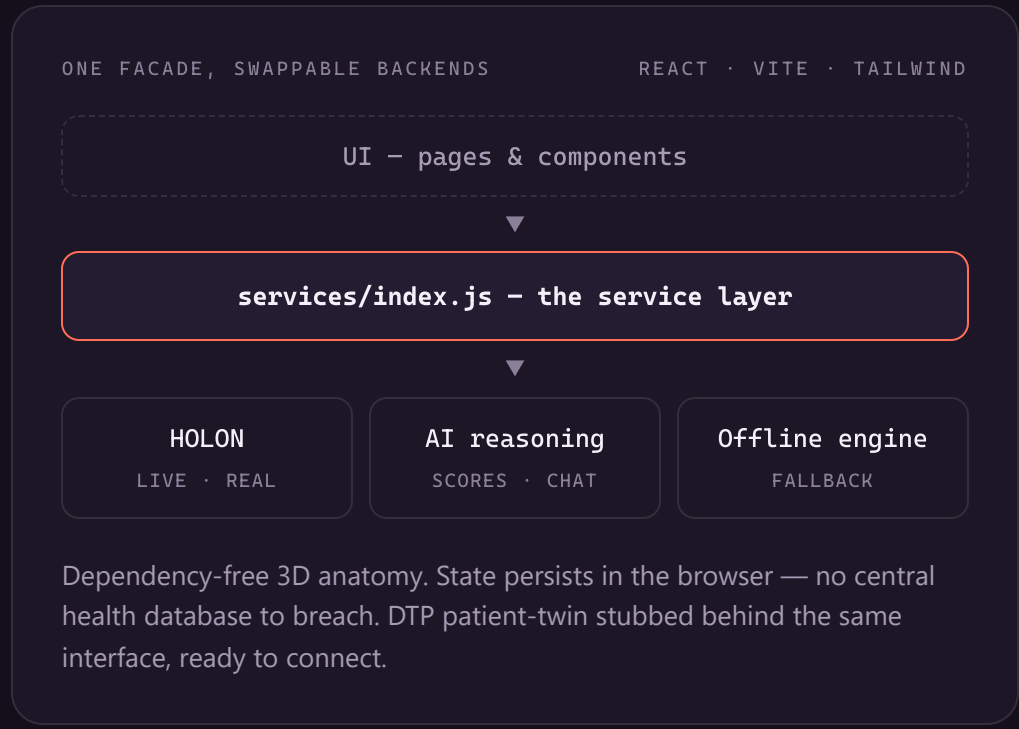


REAL HOLON, BEHIND ONE CLEAN LAYER

Wired in — and it never dead-ends.

- Two live calls on **@ontomorph/holon-client**: `concepts.search()` resolves every substance; `interactions.checkList()` screens the whole list.
- Thresholds grounded in **NHS · FDA · EFSA** guidance — every flag carries its source and evidence grade.
- Educational tool, not a medical device: **consent gate, disclaimers, emergency routing**.

SAY IT STRAIGHT Real HOLON lives behind the service layer. With no key or a flaky network it falls back to a **deterministic clinical-rule engine** — set `VITE_HOLON_API_KEY` and it goes live, no code change.



Dependency-free 3D anatomy. State persists in the browser — no central health database to breach. DTP patient-twin stubbed behind the same interface, ready to connect.

BUSINESS MODEL & WHY IT WINS

Free for students. Bought by campus health.

B2B2C through campuses: the consumer tool is the wedge, the university health service is the buyer — it carries the cost and risk of student self-medication. A sharp, underserved beachhead before anyone managing several substances at once.

Roadmap → Live DTP patient twin · Wearables · Wider HOLON coverage · Clinician view

INNOVATION

A body that reacts

Two projected futures you can watch.

CLINICAL VALUE

A real, common harm

Genuine interaction check + path to care.

EXECUTION

Works end to end

Safety framework + graceful fallbacks.

PLATFORM USE

All five pieces

HOLON, 3D, AI, twin, simulation — one layer.